

Advanced Econometric Methods:sample

Student ID: Your Student ID

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Question 1

1.1 Scatter plot

The simulated data show a positive association between X and Y. The line is the least-squares fit.

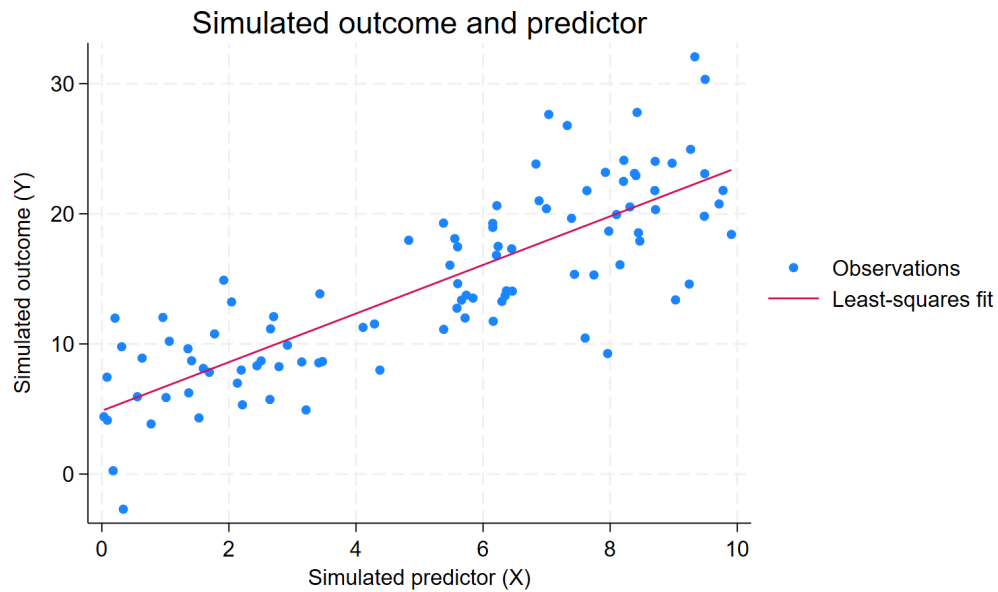


Figure 1. Simulated outcome (Y) and predictor (X).

1.2 Event-study

Treatment is assigned to 60 of 120 units. Outcomes include unit and time effects, a rising treatment effect after period 0, and random noise. Period -1 is the reference period. The plot shows two-way fixed-effects estimates and 95% confidence intervals; pre-treatment estimates should be near zero.

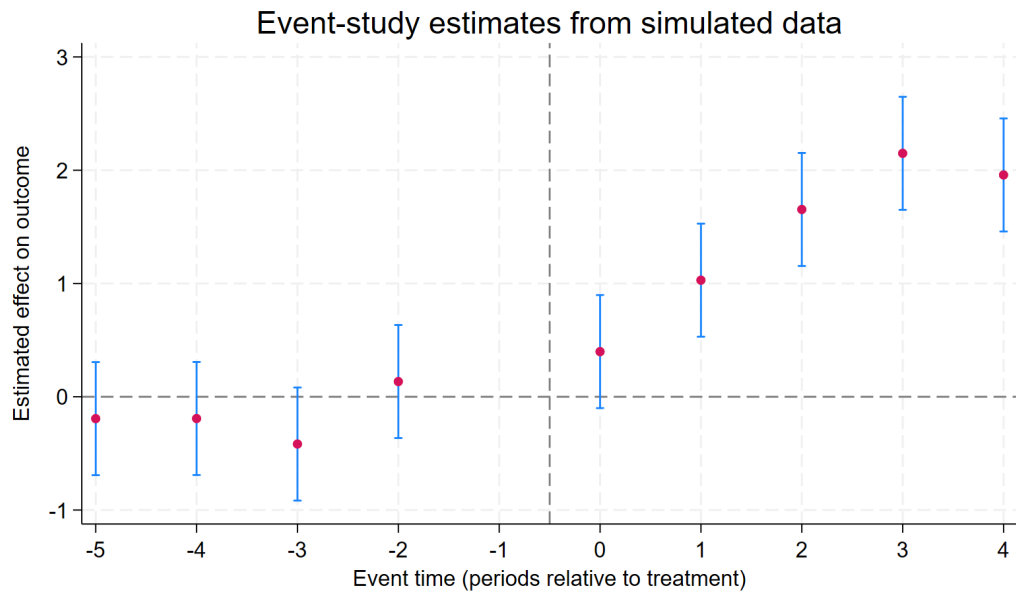


Figure 2. Event-study estimates from simulated data.

Question 2

2.1 Penguin figure

Palmer Penguins measurements by sex and species. Records with missing sex or plotted measurements are omitted.

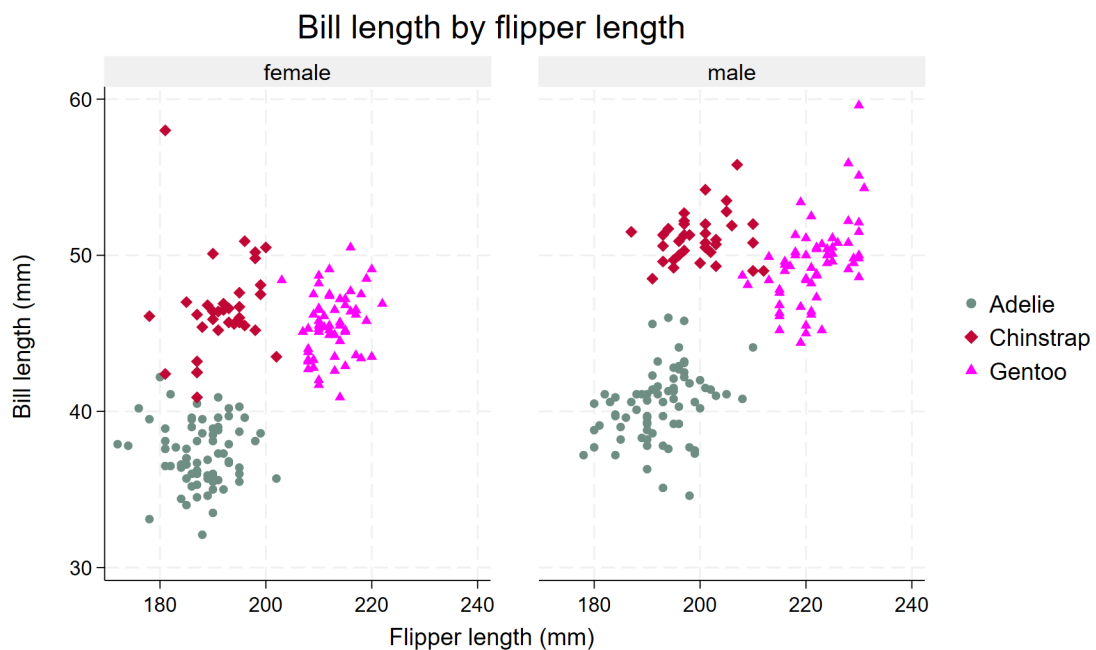


Figure 3. Penguin measurements. Source: Horst, Hill, and Gorman (2020).

2.2 Table

Mean bill length, bill depth, flipper length, and body mass by species and sex. Each mean uses available observations for that measurement.

Table 1. Penguin summary statistics by species and sex.

Species	Sex	Bill length (mm)	Bill depth (mm)	Flipper length (mm)	Body mass (g)
Adelie	Female	37.3	17.6	187.8	3368.8
Adelie	Male	40.4	19.1	192.4	4043.5
Chinstrap	Female	46.6	17.6	191.7	3527.2
Chinstrap	Male	51.1	19.3	199.9	3939.0
Gentoo	Female	45.6	14.2	212.7	4679.7
Gentoo	Male	49.5	15.7	221.5	5484.8

2.3 Regression

We estimate an OLS model of bill length on species, sex, and flipper length. The coefficients are conditional associations, not causal effects. Adelie and female penguins are the reference categories.

$$\begin{aligned}\text{Bill length}_i = & \beta_0 + \beta_1 1\{\text{Chinstrap}_i\} + \beta_2 1\{\text{Gentoo}_i\} \\ & + \beta_3 1\{\text{Male}_i\} + \beta_4 \text{Flipper length}_i + \varepsilon_i\end{aligned}$$

Table 2. OLS estimates of bill length (mm).

Bill length (mm)		
Predictor	Coefficient	Std. error
Chinstrap (reference: Adelie)	9.434***	0.354
Gentoo (reference: Adelie)	5.974***	0.650
Male (reference: female)	3.004***	0.289
Flipper length (mm)	0.101***	0.022
Constant	18.179***	4.060

Outcome mean: 43.99 mm; N: 333; R-squared: 0.832. Complete cases; conventional OLS standard errors in parentheses.

*, **, and *** indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Source: Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data (R package version 0.1.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.3960218>